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***n*-Butanol Synthesis from Ethanol on Hydroxyapatite**

Takashi TSUCHIDA¹, Shuji SAKUMA¹, Tatsuya TAKEGUCHI², Wataru UEDA²

¹Central Research Center, SANGI Co., Ltd. Fudoanno 2745-1, Kasukabe 344-0001, Japan

²Catalysis Research Center, Hokkaido University, Kita 21 Nishi 10, Kita-ku, Sapporo 001-0021, Japan

Abstract

n-Butanol was directly synthesized from ethanol on nonstoichiometric hydroxyapatite ($\text{Ca}_{10-Z}(\text{HPO}_4)_Z(\text{PO}_4)_{6-Z}(\text{OH})_{2-Z} \cdot n \text{H}_2\text{O}$; $0 < Z \leq 1$, $n = 0 \sim 2.5$: HAP). The selectivity and the yield of *n*-butanol strongly depended on contact time. The hydroxyapatite catalyzed selective ethanol conversion to *n*-butanol with maximum selectivity of 76%. When this is assumed to be a second-order reaction, the simulation plot of the reaction fits the experimental data well.

1. INTRODUCTION

It has been reported that *n*-butanol can be synthesized from ethanol on MgO [1], alkali metals/zeolite (ZSM-5)[2], $\text{MgO-CuO-MnO/silica}$ [3], Mg_yAlO_x [4] and so on. Although the elementary reactions of this process have not been probed, it is composed of a dehydration of ethanol in the overall chemical reaction formula.



In this paper, we developed highly active nonstoichiometric hydroxyapatite, and examined its ability to catalyze the conversion of ethanol to *n*-butanol and higher alcohols (C_6 -alcohol, C_8 -alcohol etc.).

2. EXPERIMENTAL

A HAP catalyst with $\text{Ca/P} = 1.64$ was prepared by the precipitation method with 0.60-mol/l of $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ and 0.40-mol/l of $(\text{NH}_4)_2\text{HPO}_4$. The catalyst was characterized by powder X-ray diffraction and measurement of its BET surface area. Ethanol conversion was carried out using an ordinary flow reactor, and products were analyzed by GC-MS and GC-FID. The ethanol conversion, yield and selectivity of products were calculated based on carbon weight.

3. RESULTS AND DISCUSSION

Hydroxyapatite with Ca/P=1.64 catalyzed selective ethanol conversion to n-butanol at 300°C and contact time 1.78 s with maximum selectivity of 76% (see Table), which is the highest value ever reported. Higher alcohols were also obtained. However the reaction mechanism has not been elucidated.

Assuming a successive reaction mechanism, in which only n-alcohol was reacted with n-alcohols, and olefins were synthesized by dehydration of alcohols and aldehydes by dehydrogenation of alcohols respectively, we simulated the ethanol conversion curve and the yield curve for n-butanol at 450°C over a range of contact times. Results are shown in the Figure. The n-butanol yield increased with an increase in contact time from 0 to 0.22 s, but decreased thereafter, and the HAP catalyst showed a maximum n-butanol yield of 19.3% at contact time of 0.22 s. The simulation plot thus fits the reaction data well.

4. CONCLUSION

Ethanol was successively converted to n-butanol and higher alcohols on HAP. We postulate this occurs by second order reaction. Contact time was found to be an important factor in increasing the yield of n-butanol

REFERENCES

- [1] A. S. Ndou, N. Plint, N. J. Coville, *Appl. Catal. A: Gen.*, **251**, (2003) 337.
- [2] C. Yang, Z. Meng, *J. Catal.*, **142**, (1993) 37.
- [3] V. Nagarajan, *Indian J. Tech.*, **9**, (1971) 380.
- [4] J.I. Di Cosimo, C.R. Apesteguia, M.J.L. Gines, E. Iglesia, *J. Catal.*, **190**, (2000) 261.

Table Effect of contact time on product selectivity from ethanol on HAP

Reaction temp (°C)	300	300	300	300
Contact time / s	0.02	0.45	0.89	1.78
Conversion (%)	0.6	4.2	10.7	14.7
Selectivity (C-wt%)				
C ₂ H ₄	13.5	2.5	0.7	0.6
CH ₃ CHO	61.7	11.5	4.4	1.7
C ₄ H ₈	2.4	2.0	1.2	0.4
CH ₂ =CH-CH=CH ₂	3.3	3.0	1.8	1.1
C ₃ H ₈ CHO	tr.	0.4	0.4	0.2
CH ₃ CH=CHCHO	tr.	0.4	0.2	0.1
C ₂ H ₅ OC ₂ H ₅	tr.	tr.	0.1	0.1
n-C ₄ H ₉ OH	3.2	61.4	75.0	76.3
C ₆ -alcohol	0.3	5.0	6.2	8.6
C ₈ -alcohol	tr.	0.3	0.5	1.0
C ₁₀ -alcohol	0.5	tr.	0.1	0.2
Aromatics	tr.	1.0	1.0	1.3
Others	15.1	12.3	8.3	8.5

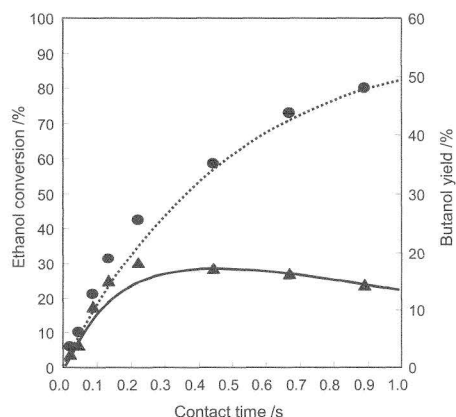


Figure Simulation curves vs. experimental data at 450°C.
Ethanol conversion is shown as ●. Yield of n-butanol is shown as ▲.